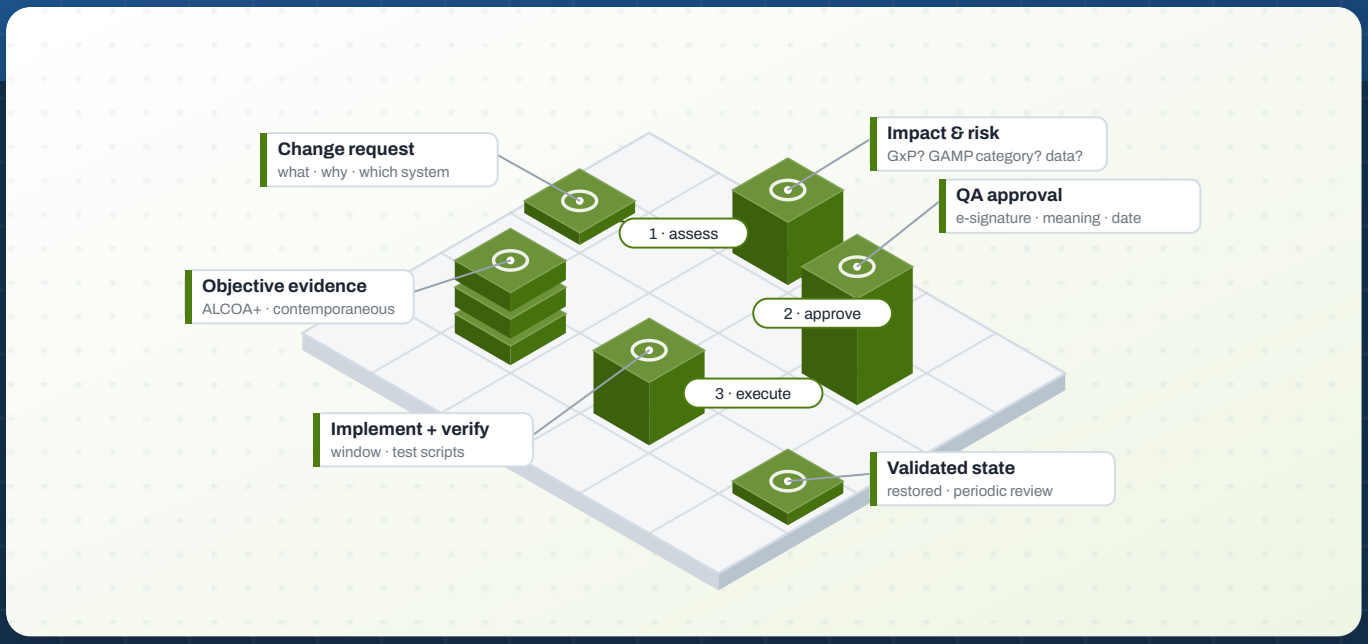
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-GOV-001

Change Control and Objective Evidence for GxP and 21 CFR Part 11 Systems

Classify a change, assess its GxP impact, plan the test and the rollback, get Quality approval with an electronic signature, implement in a window, capture objective evidence that would satisfy an inspector, and close the record without leaving the system outside its validated state.

Governance & Compliance | GxP / CSV | 21 CFR Part 11



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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain why change control protects the validated state and data integrity of a GxP system.
- Classify a change as standard, normal or emergency and perform a risk-based GxP impact assessment.
- Write a change record an auditor can follow: plan, test, rollback, approvals and evidence.
- Capture objective evidence that meets ALCOA+ and the audit-trail and e-signature expectations of 21 CFR Part 11.
- Handle emergency changes, deviations and CAPA without breaking the record, and pass the periodic review.

Time	Difficulty	Minimum access	Risk
30–60 min per record · plus test execution	Intermediate	ITSM change requester; QA approver for GxP	Compliance risk if evidence is missing

CHAPTER 1

Concepts You Need First

In a regulated laboratory or manufacturing site, a computer system that creates, stores or processes GxP records has been **validated**: someone has documented, with evidence, that it does what it is supposed to do, reliably, every time. Any change to that system, from a Windows update to a new instrument driver to a new IP address, can quietly invalidate that proof. **Change control** is the discipline that keeps the proof true: no change happens unless it is described, assessed, approved, tested and evidenced, and every step leaves a record an inspector can read years later.

KEY CONCEPT · THE VALIDATED STATE IS A PROMISE, NOT A PROPERTY

A system is not "validated" the way a server is "patched". Validation is a documented promise that the system is fit for its intended use. Change control is how you keep that promise after go-live. A change made without it does not just break a procedure; it means the organization can no longer prove that the data the system produced since then is trustworthy.

1.1 Key terms

Term	Meaning in practice
GxP	Umbrella for the "good practice" regulations: GMP (manufacturing), GLP (laboratory), GCP (clinical), GDP (distribution). A system is *GxP* if it touches records or decisions those rules govern.
CSV	Computerized System Validation: the documented process (plans, requirements, risk assessment, IQ/OQ/PQ testing, reports) that establishes fitness for intended use.

Term	Meaning in practice
CSA	Computer Software Assurance: the FDA's risk-based approach that focuses effort on features with high process risk and accepts unscripted or vendor testing where risk is lower.
Validated state	The configuration that was tested and approved. Any deviation from it must be justified, assessed and documented.
ALCOA+	Data-integrity attributes: Attributable, Legible, Contemporaneous, Original, Accurate, plus Complete, Consistent, Enduring, Available.
Objective evidence	Records that prove something happened, independent of anyone's memory: screenshots, logs, signed test steps, system reports.
Deviation	An unplanned departure from an approved procedure, test script or specification. Recorded, investigated, closed.
CAPA	Corrective and Preventive Action: the fix for a deviation's root cause, and the action that stops it happening elsewhere.

1.2 Why change control matters in GxP

Regulators do not inspect your intentions; they inspect your records. The questions an inspector asks about any change are always the same: *What changed? Who decided? On what basis? How do you know it still works? Where is the proof?* A good change record answers all five without anyone in the room having to explain it.

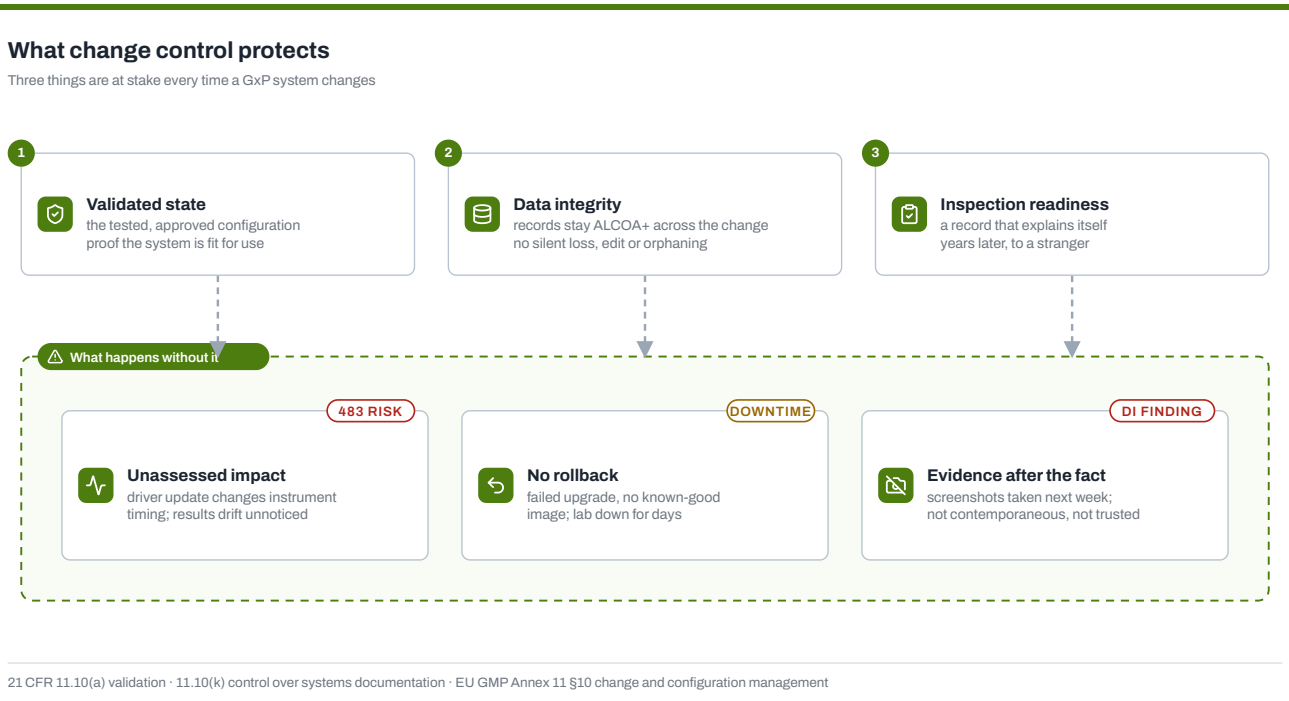


Figure 1. Change control protects three things at once. The bottom row is what inspection findings are made of.

1.3 CSV and CSA: how much testing is enough

Traditional CSV tended to script and screenshot everything. **Computer Software Assurance** keeps the same goal, confidence that the system is fit for its intended use, but scales the effort to risk. A feature that directly affects product quality or a reportable result gets rigorous, scripted testing; a feature with low process risk can be covered by unscripted, exploratory testing or by leveraging vendor evidence, documented briefly. The record still has to show *what* was tested, *by whom*, *when* and *with what result*.

> **NOTE · RISK-BASED IS NOT EVIDENCE-FREE**
>
> CSA reduces the volume of documentation that adds no assurance. It does not remove the need for an approved change, a recorded rationale for the testing approach, and evidence of the outcome. "We used a risk-based approach" is a conclusion; the record must show the risk assessment that led to it.

1.4 GAMP 5 software categories

GAMP 5 (Second Edition) groups software by how much it is configured or customized, which drives how much a change needs to be tested. Category 2 (firmware) was retired in earlier editions; firmware is now treated according to how it behaves.

Category	Description	Typical examples	Change testing emphasis
1 · Infrastructure software	Operating systems, databases, middleware, security tools	Windows 11, SQL Server, antivirus/EDR agents, backup agents	Installed correctly; dependent applications still work (regression)
3 · Non-configured	Used as supplied; run-time parameters only	Instrument control software with fixed functions, firmware-like tools	Installation, key functions for intended use
4 · Configured	Configured to the business process	LIMS, ELN, chromatography data systems, ServiceNow workflows	Configured functions, workflows, security roles, audit trail
5 · Custom	Bespoke code or macros	In-house scripts, custom interfaces, Excel macros that calculate results	Full lifecycle: requirements, code review, functional and regression testing

CHAPTER 2

Access, Prerequisites and Planning

2.1 Roles and responsibilities

Role	Responsibility in the record	Typical access
Change requester	Describes what and why; links the affected configuration items (CIs)	ITSM requester
Change owner / implementer	Writes the plan, test approach and rollback; implements; captures evidence	ITSM change owner; admin rights on the system for the window only
System owner (business)	Confirms business impact and acceptance; schedules the window with the lab	ITSM approver
QA / CSV reviewer	Confirms GxP impact assessment, test approach and evidence; approves with e-signature	ITSM approver with Part 11-compliant signature
CAB	Reviews normal changes for technical risk and conflicts	ITSM CAB group
Independent tester (where required)	Executes scripted test steps; signs each step	Read access to evidence store; test account

GOOD PRACTICE · SEGREGATION OF DUTIES

The person who implements a GxP change should not be the only person who approves it, and should not approve their own test results. If the team is small, document the compensating control (for example, QA reviews the evidence before closure).

2.2 Prerequisites checklist

- [ ] The affected system is in the CMDB with its GxP classification, GAMP category, system owner and validation documents linked.
- [ ] The current validated configuration (baseline) is known: OS build, application version, drivers, network settings, accounts.
- [ ] A rollback route exists and has been confirmed: known-good image, snapshot, backup, or documented reinstall media and settings.
- [ ] The lab or business has agreed a maintenance window, and instruments can be idle or are released from use.
- [ ] Test approach agreed with QA: which requirements are re-tested, scripted or unscripted, who executes, who reviews.
- [ ] Evidence location prepared: a controlled folder or the ITSM attachment area, with the naming convention in section 6.2.

- [ ] Related tickets raised and linked (for example domain removal, reimage, network change, decommission).

2.3 Change classification

Every change falls into one of three classes. The class decides the approval path, not the amount of care.

Class	Definition	Approval	Examples
Standard	Pre-approved, low-risk, repeatable, with a documented procedure and a template	Pre-approved by CAB/QA once; each instance logged	Monthly OS patch to non-GxP PCs via the approved ring; password reset; adding a user to an approved role
Normal	Anything not pre-approved. Assessed and approved before implementation	Change owner › system owner › CAB › QA (if GxP impact)	OS upgrade of a lab workstation; new instrument driver; static IP change; agent install
Emergency	Needed now to restore service or close an actively exploited risk	Emergency CAB verbal/expedited approval; retrospective QA approval within the agreed period	Isolating a workstation with active malware; restoring a failed acquisition PC from image

2.4 Risk and planning notes

Action	Risk	Notes
Raise and assess the change	READ-ONLY	No system impact. Do it before anything else.
Take a pre-change baseline (screenshots, config export)	LOW	Evidence of the starting state; needed for rollback and comparison.
Implement a standard change	LOW	Within the pre-approved procedure only. Anything different makes it normal.
Implement a normal GxP change	MODERATE	Only after QA approval is signed; inside the window.
Implement before approval	HIGH	Unapproved change = deviation. Never, except the emergency path.
Close without evidence review	HIGH	Common inspection finding. QA must see the evidence, not a statement that it exists.

CHAPTER 3

The Change Lifecycle

The lifecycle is the same in any ITSM tool. The screens below use ServiceNow terms (Change Request, CI, CAB, approvals, Post-Implementation Review), but the stages map directly to other platforms.



Figure 2. Every GxP change passes the same eight stages. The red gate between planning and implementation is the one inspectors check first.

3.1 Stage 1 · Request

The request answers *what* and *why* in plain language and links the configuration items. A reviewer who has never seen the system should understand the change from the short description and justification alone.

Field	Good example	Weak example
Short description	Upgrade lab workstation LAB-WS-014 (test frame control) from Windows 7 to Windows 11 Enterprise	Upgrade PC
Justification	Windows 7 is out of support; no security updates. Upgrade restores a supported, patchable OS for a GxP-relevant system.	Old OS
Configuration items	LAB-WS-014; instrument control application v3.x; test-frame controller; lab VLAN	(blank)

Field	Good example	Weak example
Linked records	Domain removal REQ, reimage REQ, network REQ, previous validation report	(none)

3.2 Stage 2 · GxP impact and risk assessment

Answer the screening questions first. If every answer is *No*, the change has no GxP impact and follows normal IT change control; record the answers anyway, because "no impact" is itself a decision an auditor may question.

1. Does the system create, modify, store, transmit or report GxP records or results?
2. Could the change affect the accuracy, completeness or availability of those records (including audit trails)?
3. Could it affect a function covered by validation (requirements, configuration specification, test scripts)?
4. Does it change security, user access, electronic signatures, time synchronization or backups of the system?
5. Does it change the OS, a driver, firmware or anything else that sits under the validated application (GAMP Category 1 impact)?

If any answer is *Yes*, rate the risk with the three factors most quality systems use, and let the result drive the test approach.

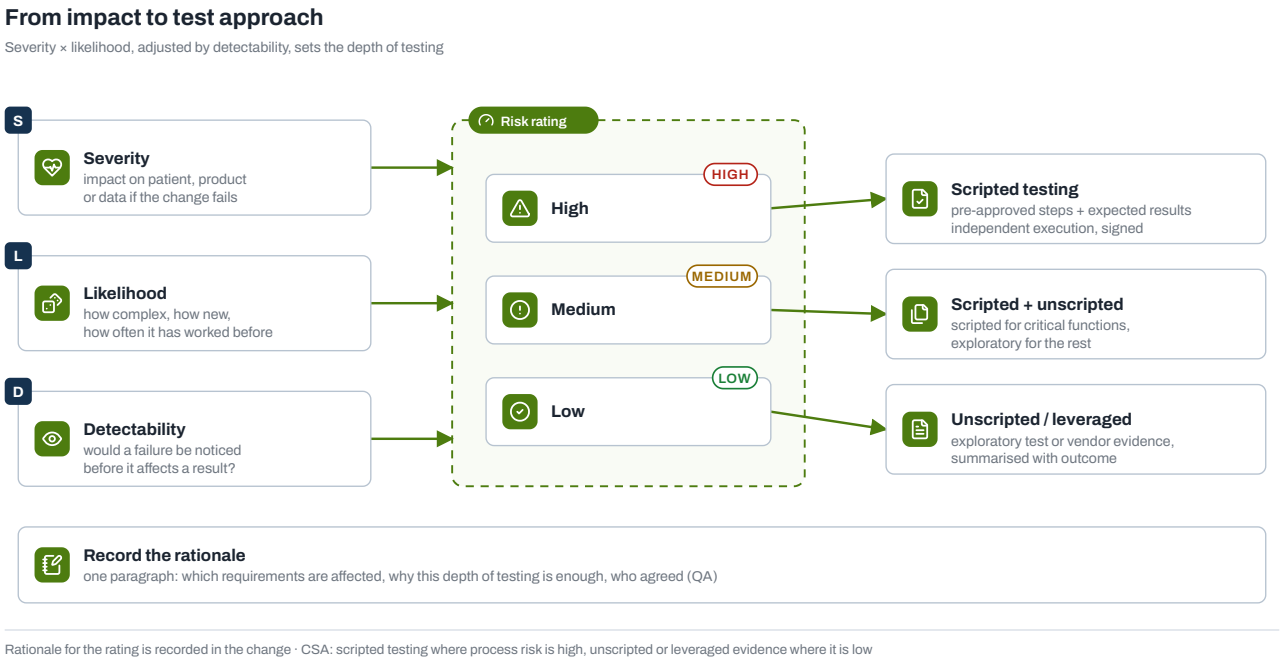


Figure 3. The risk rating is not the deliverable; the test approach and its written rationale are.

3.3 Stage 3 · Plan, test and rollback

The implementation plan is written so that someone other than the author could execute it. Each step has an expected result. The rollback plan names the trigger (what result means "stop"), the steps, and how long it takes, and it is tested or at least walked through before the window.

- [ ] Numbered implementation steps with expected results and the evidence each step produces.
- [ ] Pre-change baseline capture (configuration export, screenshots of versions, network settings, installed software).
- [ ] Test approach and the list of test scripts or unscripted test charters, with requirement traceability.
- [ ] Rollback trigger, rollback steps, time to roll back, and who decides.
- [ ] Communications: who is told before, during and after; how the lab confirms the system is back in use.

3.4 Stage 4 · Approval with electronic signature

In a Part 11 context, an approval in the ITSM tool is an **electronic signature** if it is used to sign a GxP record. The tool and the process must then meet the signature requirements below; if they cannot, the signed approval is captured in the validated quality system and referenced from the change.

Requirement	What it means for an approval
§11.50 Signature manifestations	The signed record shows the signer's printed name, the date and time of signing, and the meaning of the signature (for example *Approved for implementation*, *Reviewed evidence*).
§11.70 Signature/record linking	The signature is linked to its record so it cannot be removed, copied or transferred to falsify another record.
§11.100 General requirements	Each signature is unique to one individual and never reused or reassigned; identity is verified before the signature is issued.
§11.200 Components and controls	Non-biometric signatures use at least two distinct components (for example user ID and password); re-authentication at signing.

WATCH OUT · APPROVAL BY EMAIL IS NOT AN APPROVAL

An email saying "approved" pasted into a work note is not a controlled signature: it is not linked to the record, does not show the meaning, and can be forged. Use the tool's approval mechanism, or the validated quality system.

3.5 Stages 5–8 · Implement, verify, review, close

STEP 1 **Implement and verify inside the window**

1. Confirm the approval is signed and the window has started. Record the actual start time in the change.

2. Capture the pre-change baseline evidence exactly as planned (section 6.2 naming).
3. Execute the implementation steps in order. Record each step's actual result next to the expected result.
4. If any step does not meet its expected result, stop. Decide with the system owner: fix within the approved plan, or roll back. Anything outside the plan is a **deviation**.
5. Execute the test scripts or unscripted tests. Testers sign and date each step at the time it is executed.
6. Return the system to use only after the system owner confirms. Record the actual end time.

STEP 2 Post-implementation review and closure

1. Complete the Post-Implementation Review: did the change achieve its objective, were there deviations or incidents, was the rollback needed?
2. Attach all evidence to the record; check every planned evidence item is present and legible.
3. Update the CMDB and the validated configuration documents (configuration specification, inventory, SOP references).
4. QA reviews the evidence (not a summary of it) and signs closure with the meaning *Evidence reviewed; system in validated state*.

CHAPTER 4

Test Evidence and Qualification

4.1 IQ, OQ and PQ

Qualification stages come from equipment and system validation. For infrastructure changes under a validated application, the change usually re-executes a subset of each.

Stage	Question it answers	Evidence for a workstation OS upgrade
IQ · Installation Qualification	Is it installed as specified?	OS edition and build; application and driver versions; hostname; domain or workgroup membership; time source; security agents
OQ · Operational Qualification	Does it operate as specified across its range?	Application starts; communicates with the instrument; user roles enforced; audit trail records actions; data saves to the right location
PQ · Performance Qualification	Does it perform in real use?	A representative run (for example a test frame cycle) produces results matching the reference system within tolerance

4.2 Scripted versus unscripted testing

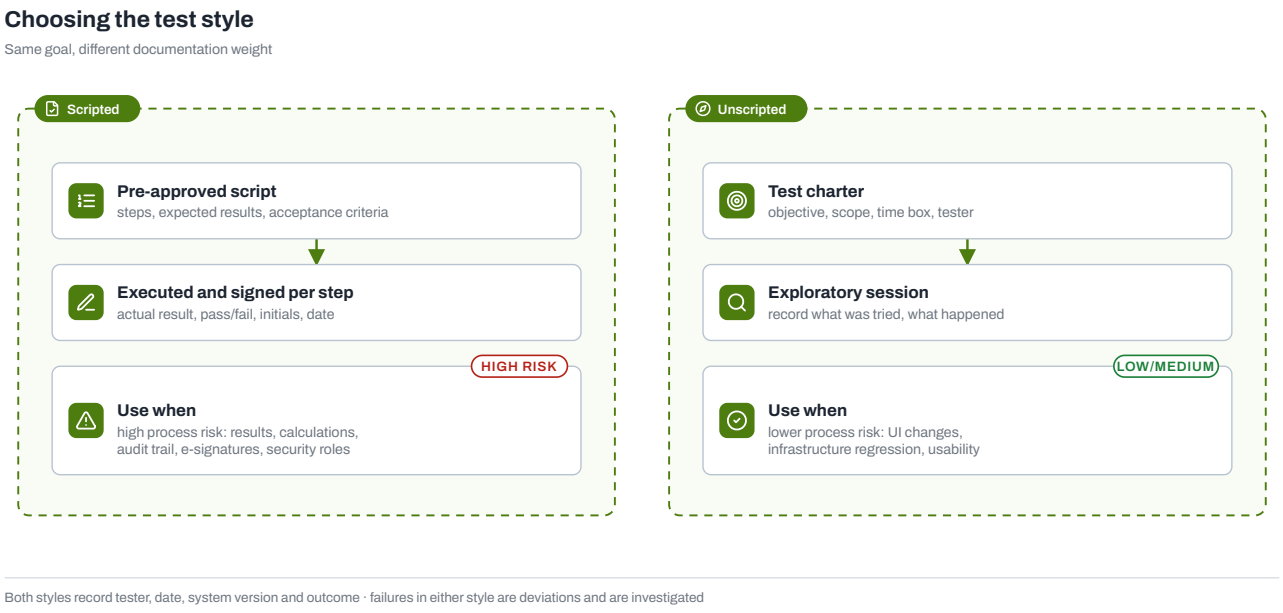


Figure 4. Scripted testing for what could change a result; unscripted testing for confidence everywhere else.

4.3 Writing a test step that will survive review

Element	Example
Step	Launch the instrument control application from the Start menu as a standard lab user.
Expected result	Application opens without errors; *Help* › *About* shows version 3.x.y; status bar shows *Controller connected*.
Actual result	As expected. Version 3.x.y. Controller connected at 10:42.
Pass / fail	Pass
Evidence reference	`CHG0012345_OQ-03_LAB-WS-014_20261012-1042.png`
Executed by / date	Initials and date written at the time of execution (or e-signed in the test tool).

CHAPTER 5

Objective Evidence and Data Integrity

Evidence is objective when a stranger can look at it and reach the same conclusion you did, without asking you anything. It is trustworthy when it meets ALCOA+.

5.1 ALCOA+ applied to change evidence

Attribute	For a screenshot or log	Common failure
Attributable	Shows or is signed by the person who captured it; the system is identified (hostname, asset tag)	Screenshot with no hostname or user
Legible	Readable at normal zoom; text not cropped; annotations do not hide data	Thumbnail-size images; cropped version numbers
Contemporaneous	Captured at the time of the step; system clock visible	Screenshots taken days later "to show what it looked like"
Original	The first capture, or a verified true copy; not retyped	Version number typed into the ticket instead of captured
Accurate	Matches what actually happened, including failures	Failed first attempt omitted
Complete	All planned evidence present; nothing deleted	Rollback test evidence missing
Consistent	Same naming, time zone and order throughout	Mixed time zones; shuffled file names
Enduring	Stored in the controlled record for the retention period	Kept in an engineer's Downloads folder
Available	Retrievable for review and inspection	Attached to a personal email

5.2 Evidence standards

Every screenshot used as GxP evidence shows, in the frame or in an accompanying system report: the **hostname** (or asset tag), the **date and time** (taskbar clock or `Get-Date` output), the **user** context, and the **system or software version** relevant to the step. Use the naming convention below so files sort in execution order.

NAMING CONVENTION TEXT

```
<CHANGE>_<STAGE-STEP>_<HOSTNAME>_<YYYYMMDD-HHMM>.<ext>
CHG0012345_PRE-01_LAB-WS-014_20261012-0915.png      # baseline: winver + hostname
CHG0012345_IQ-04_LAB-WS-014_20261012-1130.png       # driver version in Device Manager
CHG0012345_OQ-03_LAB-WS-014_20261012-1042.png       # application connected to controller
CHG0012345_ROLLBACK-TEST_LAB-WS-015_20261008-1400.png
```

A short script makes the identifying details part of every capture. Run it in a window that stays visible beside the application while screenshots are taken, and save its output as its own evidence file.

EVIDENCE HEADER · RUN BEFORE EACH CAPTURE SET — POWERSHELL

```powershell
# Prints the identifying details an inspector looks for; keep the window in frame
$os = Get-CimInstance Win32_OperatingSystem
[pscustomobject]@{
  Hostname   = $env:COMPUTERNAME
  User       = "$env:USERDOMAIN\$env:USERNAME"
  DateTime   = (Get-Date -Format "yyyy-MM-dd HH:mm:ss K")
  OS         = "$($os.Caption) $($os.Version) build $($os.BuildNumber)"
  TimeSource = (w32tm /query /source)
} | Format-List

# Installed software inventory as a dated, attributable text record
Get-ItemProperty HKLM:\Software\Microsoft\Windows\CurrentVersion\Uninstall\*,
                 HKLM:\Software\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\* |
  Where-Object DisplayName | Sort-Object DisplayName |
  Select-Object DisplayName, DisplayVersion, Publisher |
  Export-Csv "C:\Evidence\CHG0012345_IQ-01_$env:COMPUTERNAME_$(Get-Date -f yyyyMMdd-HHmm).csv" -
NoTypeInformation
```

WATCH OUT · THE CLOCK IS PART OF THE EVIDENCE

If the workstation's clock is wrong, every timestamp on every screenshot is wrong. Confirm the time source (`w32tm /query /status`) in the pre-change baseline. A workgroup PC removed from the domain may lose its time source; configure NTP as part of the change.

5.3 Audit trails

21 CFR 11.10(e) requires secure, computer-generated, time-stamped audit trails that independently record the date and time of operator entries and actions that create, modify or delete electronic records, without obscuring previously recorded information, retained at least as long as the records themselves. For change control this has two consequences:

- The **ITSM record** itself needs an audit trail: field changes, state transitions and approvals are logged with user and time and cannot be edited away.
- The **GxP application** must keep its audit trail across the change: an OS upgrade, migration or reinstall must not reset, truncate or orphan it. Verify the audit trail as an OQ test, and confirm historical entries remain readable.

CHAPTER 6

Worked Example · Upgrading a Legacy Lab Workstation

A sanitized version of a real engagement: a lab workstation running instrument control software on Windows 7 had to move to Windows 11 without breaking communication with test-frame hardware. The technical procedure is SOP-LAB-001; this chapter shows the change control around it.

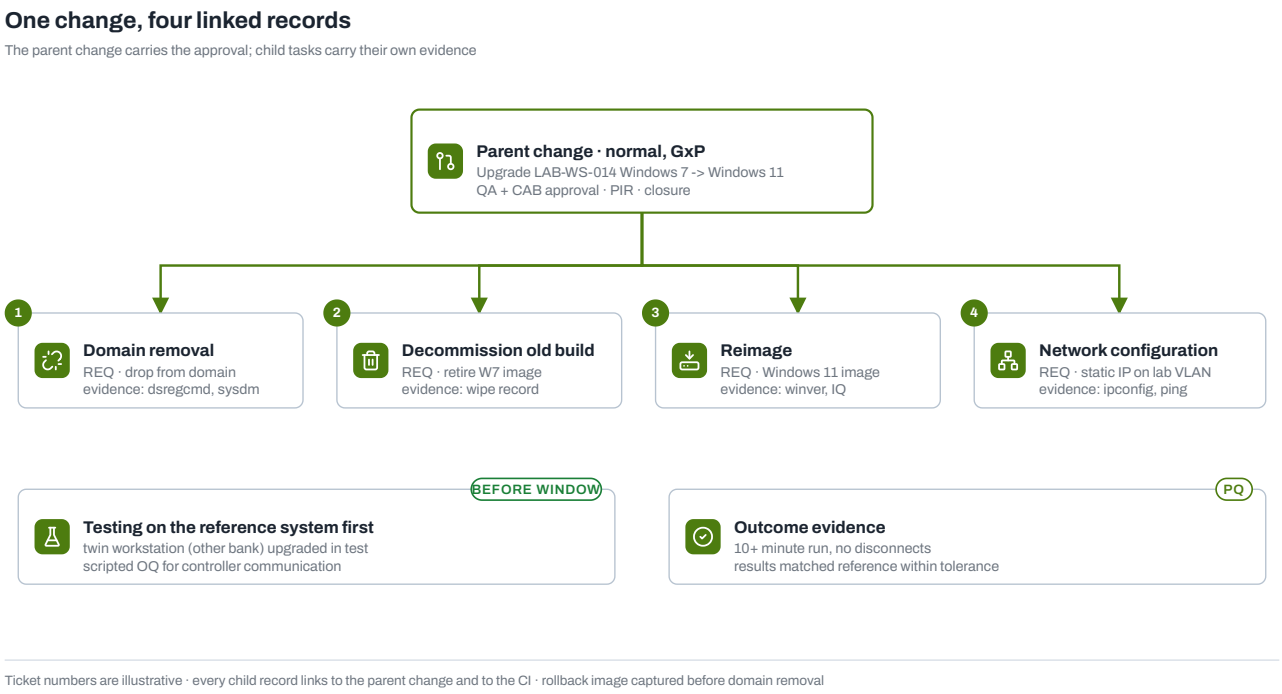


Figure 5. The worked example as change records. Splitting the work into linked child tasks keeps each piece of evidence next to the step that produced it.

Stage	What the record contained
Request	Justification (unsupported OS in a GxP area), affected CIs, link to the original validation report and the vendor's compatibility statement.
Assessment	GxP impact *Yes* (Category 1 change under a Category 3 application); severity high (test results), likelihood medium (vendor did not formally support the new OS), detectability medium.
Plan	Reference-system trial first; compatibility-mode install as a fallback; static IPv4 for the controller link; rollback = restore the captured Windows 7 image.
Approval	System owner, CAB and QA approvals with signature meaning recorded; window agreed with the lab around test schedules.
Implement	Child tasks executed in order: image capture, domain removal, reimage, application install, network configuration.

Stage	What the record contained
Verify	IQ (OS build, application version, drivers, IP settings), OQ (controller detected, audit trail, user roles), PQ (representative cycle compared with the reference bank).
PIR and close	No deviations. CMDB and configuration specification updated. QA reviewed every evidence file and closed the change.

GOOD PRACTICE · TRY IT ON THE TWIN FIRST

Where a lab has paired systems (left and right bank, or primary and backup), run the change on the one not in use and keep the other as the working reference. The comparison becomes your PQ evidence, and the reference stays available if you roll back.

CHAPTER 7

Emergency Changes, Deviations and CAPA

7.1 Emergency changes

An emergency change trades the order of steps, not their existence. Implementation may happen before full approval, but the record must still show who authorized it at the time, what was done, what evidence was captured, and a retrospective review and QA approval within the period your quality system defines (often the next business day).

STEP 1 **Emergency change path**

1. Get authorization from the emergency approver named in your process (for example the on-call change manager and the system owner). Record name, time and how it was given.
2. Take whatever baseline evidence the situation allows (at minimum a photo or screenshot of the state before you act).
3. Act to restore service or contain the risk. Record each action with a time as you go.
4. Raise or complete the change record the same day; mark it *Emergency* and link the incident.
5. Assess GxP impact retrospectively; run the verification that the risk requires; attach evidence.
6. Obtain retrospective CAB and QA approval; complete the PIR, including whether the emergency could have been avoided.

If retrospective approval is refused, the change becomes a deviation and is investigated through the deviation process.

7.2 Deviations

A deviation is raised when something departs from the approved plan, a test script, or a procedure: a step that fails, a step done out of order, an unplanned extra step, missing evidence, or work done outside the window. It is not a punishment; it is how the record stays honest.

Situation	Deviation?	Handling
A test step fails and is re-run after fixing a typo in the script	Yes	Record the failure, the cause, the correction (QA approved), and the re-run result.
An extra registry change was needed that was not in the plan	Yes	Record what and why; assess impact; QA decides whether it needs additional testing.
Screenshot missing for one IQ step	Yes	Do not recreate later and pass it off as original. Record the gap; capture current state as a new, dated record if QA agrees.

Situation	Deviation?	Handling
Window overran by 30 minutes, all steps per plan	Usually no	Record actual times; note in PIR.

7.3 CAPA



Figure 6. A deviation is closed by fixing the instance; a CAPA is closed only when the cause is gone and the fix has proved itself.

7.4 Periodic review

Validated systems are reviewed on a defined cycle (commonly every one to three years, based on risk). The review asks whether the system is still in its validated state. Change records are a primary input: an auditor will sample them.

- [ ] List of all changes since the last review, with class and GxP impact; any unapproved changes found?
- [ ] Open deviations and CAPAs related to the system; any trends?
- [ ] Configuration today versus the configuration specification; differences explained by approved changes?
- [ ] User access review completed; audit trail reviewed; backups and restores tested.
- [ ] Supplier status: OS and application still supported? (An unsupported OS is a finding waiting to happen.)

CHAPTER 8

Templates

8.1 Change record fields

Field	Required for	Notes
Number, class, state	All	Generated by the tool; state transitions audit-trailed.
Short description, justification	All	Plain language; what and why.
Configuration items	All	Linked from CMDB, including GxP flag and GAMP category.
GxP impact screening (5 questions)	All	Answers recorded even when all are *No*.
Risk rating and rationale	Normal, emergency	Severity, likelihood, detectability; one-paragraph rationale.
Implementation plan	Normal, emergency	Numbered steps with expected results.
Test approach and traceability	GxP impact	Scripts or charters; requirements re-tested; executor and reviewer.
Rollback plan	Normal, emergency	Trigger, steps, duration, decision owner.
Planned and actual window	Normal, emergency	Planned start/end; actual start/end recorded at the time.
Approvals (signature meaning)	Per class	System owner, CAB, QA; name, date/time, meaning.
Evidence attachments	GxP impact	Named per convention; listed in an evidence index.
Deviations	If any	Linked deviation records.
PIR and closure signature	Normal, emergency	Outcome, lessons, QA closure meaning.

8.2 Evidence checklist for an infrastructure change

- ☐ Pre-change: hostname, OS version and build, application and driver versions, network configuration, time source.
- ☐ Rollback point: image or backup identifier and verification that it is restorable.
- ☐ IQ: each installed or changed component and its version; security agents present and reporting.
- ☐ OQ: application launch, instrument or interface communication, user roles, audit trail entries, data storage path.

- [ ] PQ (where required): representative run compared with reference or acceptance criteria.
- [ ] Post-change: CMDB updated; configuration specification updated; system owner release to use.

CHAPTER 9

Verification and Troubleshooting

9.1 Verification checklist

Check	How	Expected
Approval before implementation	Compare approval timestamp with actual start	Approval earlier, or emergency authorization recorded
Signature meaning present	Open each approval	Name, date/time, meaning shown
Evidence complete	Evidence index vs plan	Every planned item present, named per convention
Evidence identifies system and time	Open a sample of screenshots	Hostname, date/time, user, version visible
Deviations closed or tracked	Linked records	Closed, or open with CAPA owner and due date
Configuration documents updated	CMDB and configuration specification	Reflect the new state; version incremented
Audit trail intact	Application audit trail viewer	Entries before and after the change both readable

9.2 Common audit findings and fixes

Finding	Why it happens	Fix
Change implemented before approval	Window pressure; approval stuck with an absent approver	Delegated approvers in the tool; emergency path for genuine emergencies only.
Evidence does not identify the system	Cropped screenshots	Evidence header script; template with mandatory hostname/time fields.
Screenshots dated after the change	Evidence collected at closure	Capture at each step; reviewer checks timestamps against the window.
"No GxP impact" with no rationale	Screening questions skipped	Make the five questions mandatory fields.
Rollback never tested	Assumed the image works	Restore test on a spare or the twin before the window; evidence attached.
Same person requested, implemented and approved	Small team	Independent QA evidence review before closure; documented compensating control.

Finding	Why it happens	Fix
Audit trail reset after reinstall	Application database recreated	Plan data and audit-trail migration; OQ test that historical entries remain.

CHAPTER 10

Summary, Glossary and Review

10.1 Chapter summary

- Change control keeps a validated system's promise true after go-live; without it, the organization cannot prove its data is trustworthy.
- Classify every change (standard, normal, emergency); the class sets the approval path.
- Screen for GxP impact, rate the risk, and let the risk set the depth of testing (CSA), with a written rationale.
- Approvals on GxP records are electronic signatures: name, date/time and meaning, linked to the record.
- Evidence must be ALCOA+: identify the system, the time and the person, and be captured at the time.
- Deviations keep the record honest; CAPA removes the cause; periodic review samples it all.

Check your understanding

1. An engineer upgrades a lab PC's network driver to fix a dropout and plans to raise the change afterwards. What class of change is this, and what should have happened?
2. Which three details must every evidence screenshot show, and why does the system clock matter?
3. What does §11.50 require an approval to display?
4. A test step fails because the script had the wrong expected version. Is this a deviation? How is it handled?
5. Why is "no GxP impact" still a decision that needs recording?

10.2 Glossary

CAB	Change Advisory Board: reviews normal changes for technical risk, conflicts and readiness.
CMDB / CI	Configuration Management Database and its configuration items, the systems a change affects.
GAMP 5	ISPE guidance for a risk-based approach to GxP computerized systems; defines software categories.
IQ / OQ / PQ	Installation, Operational and Performance Qualification.
PIR	Post-Implementation Review: did the change achieve its purpose without unintended effects?

Retrospective approval	Approval of an emergency change after implementation, within a defined period.
True copy	A verified copy of an original record that preserves its content and meaning.

10.3 Framework references

Framework	Controls addressed
FDA 21 CFR Part 11	§11.10(a) validation · §11.10(d) limiting system access · §11.10(e) audit trails · §11.10(k) controls over systems documentation · §11.50 signature manifestations · §11.70 signature/record linking · §11.100, §11.200 electronic signature requirements
EU GMP Annex 11	Validation · audit trails · change and configuration management · periodic evaluation · electronic signatures
NIST SP 800-53 Rev. 5	CM-3 Configuration Change Control · CM-4 Impact Analyses · CM-5 Access Restrictions for Change · SA-10 Developer Configuration Management
ISO/IEC 27001:2022	Annex A 8.32 Change management
ISPE GAMP 5 (Second Edition) · FDA CSA guidance	Risk-based assurance; software categories; scripted and unscripted testing